

Integration of temporal and semantic components into the Geographic Information through mark-up languages.

Part I: definition

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Abstract. This paper raises the issue of a research work oriented to the storage, retrieval, representation and analysis of dynamic GI, taking into account the semantic, the temporal and the spatiotemporal components. We intend to define a set of methods, rules and restrictions for the adequate integration of these components into the primary elements of the GI: theme, location, time [68]. We intend to establish and incorporate three new structures into the core of data storage by using mark-up languages: a semantic-temporal structure, a geosemantic structure, and an incremental spatiotemporal structure. The ultimate objective is the modelling and representation of the dynamic nature of geographic features, establishing mechanisms to store geometries enriched with a temporal structure (regardless of space) and a set of semantic descriptors detailing and clarifying the nature of the represented features and their temporality. Thus, data would be provided with the capability of pinpointing and expressing their own basic and temporal characteristics, enabling them to interact each other according to their context, and their time and meaning relationships that could be eventually established. All of this with the purpose of enriching GI storing and improving the spatial and temporal analyses.

Keywords: spatiotemporal reasoning, GIS, time, geosemantic, dynamic storage,

Introduction

Time and semantics are two broad, general concepts applicable to a considerable number of scenarios (physics, geology, grammar and geography, among others). Time seems to be bound to every performed action; and every one of these actions has a meaning, a sense or a way of being interpreted that may be described through semantics.

In the realm of the Geographic Information Sciences these concepts have evolved following different paths. For over 25 years, time in GIS has been an active research line [57, 59] with important theoretical and conceptual advances having been achieved. The time models developed for the temporal databases (see [69]) have chiefly influenced the trends followed for incorporation of temporal structures into

GIS. Semantics in turn derives directly from the study of language and meaning; from a computational viewpoint, these concepts are framed within the Natural Language Processing (NLP), a line of the Artificial Intelligence dealing with modelling and processing of human language in which a lot of work has been carried on for over 60 years [79, 80]. The study of meaning in the realm of Geographic Information (GI) and the Geosciences is widely related to these elements, giving way to specific working areas such as Geosemantics [56] or Geographic Information Retrieval [35].

These constructs have come together leading to initiatives such as the computational processing of temporal expressions, a research area which has aroused the interest of the academic community, as witnessed by the related multiple academic events which have been held [5, 7, 12, 13, 28, 31, 52]. In this regard relevant advances have been made: initiatives for the generation of temporal mark-up standards [25], annotation tools and systems [29, 31, 48], temporal annotation corpora [66, 78], annotation languages [72, 76, 77] and assessment methods [9, 81], most of this work having been headed by Dr. James Pustejovsky [27].

The analysis of temporal expressions allows placing data, facts and events on timelines subjectively, correlating and arranging them chronologically. With a plain temporal description there would be sufficient available elements to solve elementary questions such as When do events occur? How often are they updated? Or which one has occurred before or afterwards? In order to provide data with the ability to describe themselves and relate to one another, taking into account temporality criteria and their representation meaning, it would be necessary to enrich them with a structure enabling identification of their temporal characteristics, their context and their meaning. Such a structure should be computationally usable and should be based on standards to ensure the interoperability of the data enriched with semantic and temporal elements.

This paper raises the issue of a research work oriented to the storage, retrieval and representation of dynamic GI, taking into account the semantic, the temporal and the spatiotemporal components. We intend to establish and incorporate three new structures into the core of data storage using mark-up languages: a semantic-temporal structure, a geosemantic structure, and an incremental spatiotemporal structure. The ultimate objective is the modelling and representation of the dynamics nature of geographic features (which are dynamic by definition), establishing mechanisms to store geometries enriched with a temporal structure and a set of semantic descriptors detailing and clarifying the nature of the represented features and their temporality.

A result of this proposal would be the definition of a new storing format which unlike the current model, would be made up of the three basic components (attributes, location and temporal reference) in addition to three layers to incorporate the semantic and temporal components, these layers are the core and innovation of this proposal (see Fig.1). Initially a file-type storing format is proposed; besides, the possibility of integrating the new concepts into a spatial database engine will be assessed according to the degree of progress of the research work, and we will analyse advantages, drawbacks and feasibility.

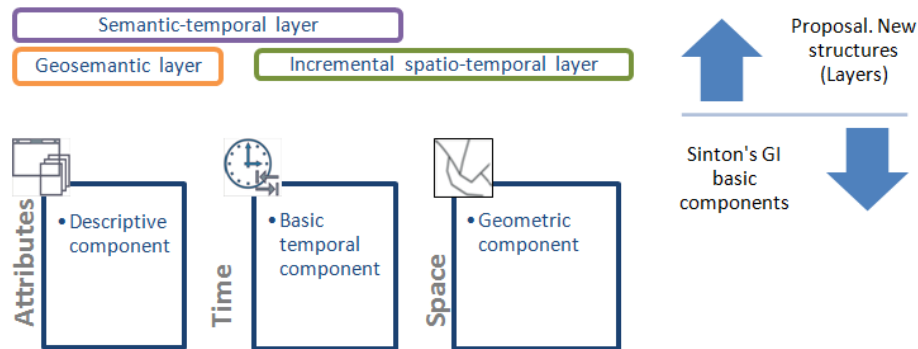


Fig. 1. Basic components of the GI and new structures proposed

This piece of work is primarily based on: the space and time concepts developed by Langran [40, 43], Peuquet [58-64], Worboys [84, 85, 87-89], Hornsby and Yuan [24, 93-96, 98-100], Galton [20, 23] and Wachowicz [83]; the semantic and analysis concepts concerning the annotation of temporal expressions as developed by the Brandeis University [5-7]; the TimeML Standard [25, 81]; the work related to space and time labelling developed at the MITRE Corporation [9, 72]; as well as the concepts of semantics and GI retrieval proposed by Jones [33-35], Manning [45] and Markowetz [47].

This paper is structured as follows; part two presents basic and related concepts of time and geosemantic as well as the initial point of this research. Part three shows the problem that this research is intended to solve, it presents current models and some pros and cons. Part four describes in detail the proposal and the research scope by giving the hypothesis, objectives, and the metamodel elements. It also shows the new components and how these components will be included into the GI. Finally, part six enumerates the conclusion and the anticipated contributions to be gained from this research.

Relevant concepts, grounds and starting point

Both time and space appear to be indivisible or at least deeply related to one another; they make up an abstract universal frame, and everything surrounding us is contained therein. In this regard, every geographic feature and geographic entity belongs to that frame, therefore also its characteristics, properties and all possible analyses developed on it. In this context, Couclelis [10] describes what the spatiotemporal analysis and reasoning is:

“Geographic entities, like everything else in the world, exist in time as well as in space;..... Spatio-temporal reasoning is not reasoning about some abstract (x,y,z,t) framework: it is mainly reasoning about the appearance, change, and disappearance of things in space and over time.”

For decades, Geography has emphasised the spatial component, leaving aside the temporal component. From its inception, the GIS development focused on the analysis of the geographic elements (space); this characteristic was inherited and the temporal aspect was consigned to the second place. By following the quantitative revolution of the 60's, Geography was catalogued as the "space science", so that the temporal component was limited to a simple attribute [57]. One of the first references to spatiotemporal data is to be found in the study published by Donna Peuquet [59], who by reflecting on temporal data series, identified the nature and the importance of the temporal dimension setting it apart from the spatial dimension.

Such as Ott and Swiaczny indicate [57], time is a relevant aspect in the subject field of GIScience, an element attracting ever so much attention by its importance, particularly during the last decade. This might be due to the large volume of data we now have at our disposal and to the potential that the historical series of data have generated¹, whereby the lack of effective methods and tools to deal with these data is shown; there are not methods to process and analyse temporal data appropriately. The time variable provides an additional component in GI management and it needs to be dealt with differently using new methods and models.

Concepts about time

Regarding time, it is necessary to define it, to be aware of the type we are talking about and the model describing it: mechanical, mathematical, organic or psychological. How to perceive and model time is a mainstay of this research; to that end we are starting from the pre-established models, accepting their premises, theorems, restrictions and additional characteristics. In this research study, the mechanical and mathematical models will be taken on, with which all geographic features perceived by our senses (rivers, ways) or abstractly modelled (airways) may be described from a temporal viewpoint. With these two types of models it is possible to define elements such as measure, interval, dimension, modification, instant and position among others.

Regarding the form of relating space and time, two schools of thought have evolved from the ancient Greece [57]: (i) the absolute perspective where space and time are perceived as a single container within which everything exists, and (ii) the relativist perspective, where space and time are perceived as interrelated, dependent elements. In this research project the applied perspective will be the Newtonian one, where time is envisioned as an independent dimension though widely related and similar to the spatial dimension. This approach is used in view of the flexibility it provides for independent handling of the temporal component.

Castagneri [32] defines the temporal GIS (time-integrative GIS) as a means of storing and analysing spatial objects and changes in their attributes over time. This

¹ On this matter Peuquet [64] states that there is a large amount of spatiotemporal data at different scales, geographically taking up wide zones covered by satellite images captured by specialized sensors or random elements captured by mobile devices, and temporally including collections gathered during decades or datasets captured in real time.

definition allows identifying the three types of elements (data) proposed by Peuquet as the basic elements of a GIS incorporating the time variable [62]: space, time, and attributes, basic components of the triadic model defined by Sinton [68] (see Fig. 2 and Fig. 3). The three main components of GI have been successfully and independently analyzed for over four decades; instances of these analyses and the definition of their primary elements are the studies of Berry [4] and Sinton. Peuquet's concept would be reinforced later on by Openshaw [55] through the analysis of patterns in the triad space-time-attribute. The triadic model surmises the representation of the attributes of an object with a specific position at a certain time, and like Openshaw's work, it shows the existing relationship between the where, the when and the what.

Nevertheless, it should be noted that these studies were oriented toward the structure and storage in the databases to be used by the GIS and that the concept of analysis and exploitation was still fuzzy; the interoperability concept was not taken into account either. The whole and integrated analysis of the main components is still a pending task; the proposed models so far are not entirely satisfactory and they do not provide a solution to the current needs for analysis and extraction of information.

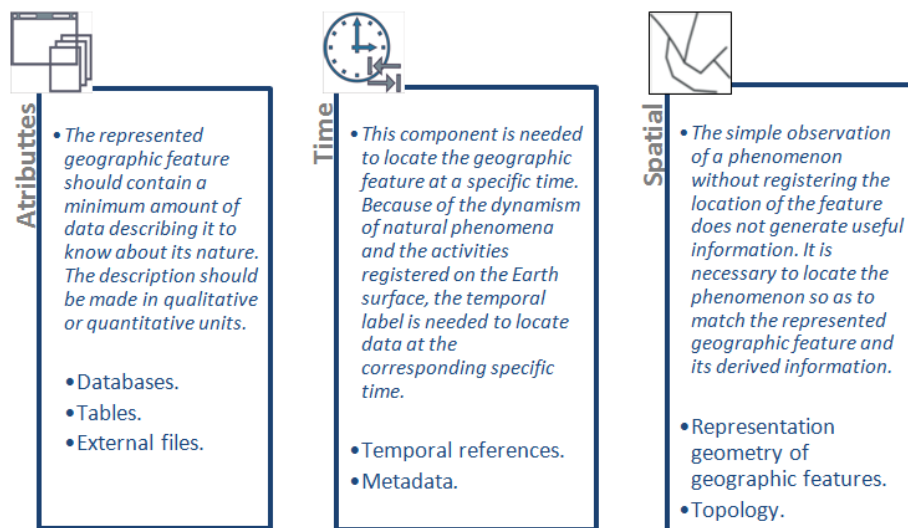


Fig. 2. Description of the three basic components of Geographic Information defined by Sinton

Ott and Swiaczny [57] remark that the chief purpose of a temporal GIS is to reproduce temporal processes or sequences of events of the real world in a model, so as to make them accessible for the realisation of spatial queries, analysis and visualisation. However, in order to present and analyse the temporal and spatiotemporal information and their relationships, the modelling process requires an absolute accuracy in all the characteristics of the modelled, stored data: geometry, temporality, cartographic projection and attributes, among others.

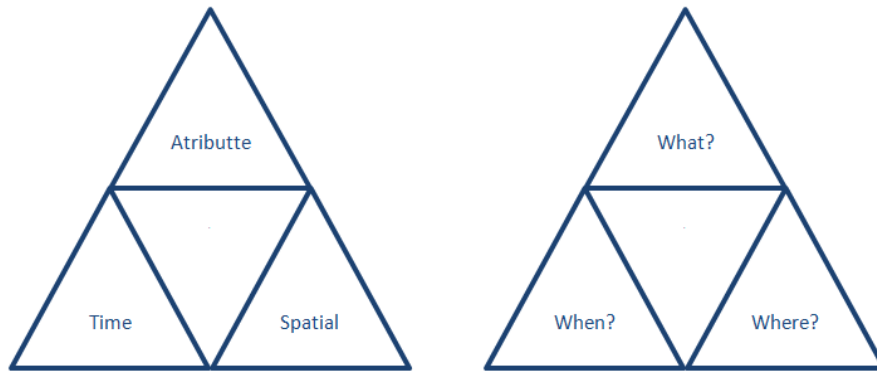


Fig. 3. Triadic model of space, time and attributes (Peuquet) [62]

The main issue in a historical or temporal GIS is the need to count on a high degree of reliability in their two basic components, spatial and temporal. It is possible to reach that degree of reliability only if the problem is tackled from the very beginning, at the start of the process, this is from the modelling of both the data and the storage. So far many applications have been carried out which solve specific problems; then they are used in generic scenarios with the purpose of creating frameworks for the treatment of spatiotemporal data. Nevertheless, when working with models intended for specific settings or when trying to develop dynamic tasks with tools designed for static scenarios and data, it becomes patently clear that those models cannot be applied to general aspects of the geographic features; they cannot be applied to the dynamic environment containing them either.

Peuquet [60, 65], Langran [39, 42-44], Ott and Swiaczny [57], have stressed the need to work with space and time in GIS in a correlated fashion. By analysing the traditional way in which GIS are modelled by layers, it becomes apparent that the thematic and temporal levels are separated in different categories. This is due on the one hand to the above mentioned issue of the preference of space over time and on the other hand because the current models gear (almost compel) the user to work in a predetermined way with the existing spatial tools, having to forego the temporal component. Many free, open, and proprietary GI management software handle vector formats where the geometric component displaces the temporal component; this is due to the form in which data are structured and stored in the different types of formats and/or store methods; as an example, the old Shapefile format [16] stores geometry and attributes representing and describing geometric entities, but temporality and evolution of the displayed geographic feature over time is omitted. More recent, advanced formats such as the Geodatabase [15], in addition to storing geometry and attributes, incorporate characteristics and topological relationships, spatial integrity, versioning and an option of historical file of data based on replication; however none of them carries these functions in an incremental manner and without geometry duplication.

Geosemantic and temporal analysis

Regarding the semantic and geosemantic components, multiple studies [3, 8, 33, 34] show how the temporal and spatial components play an important role by filtering, gathering together and prioritising information resources, thus motivating research in methods that would allow turning geotemporal references described by humans in structures and representations understandable by computers and digital information processing systems.

The automatic retrieval of geotemporal information is related to (i) the processes of identification and text analysis, (ii) the search of elements and geographic and temporal references and their accurate location on the space, and (iii) the combination of these references in meaningful semantic compilations such as the indicated search context [67]. The analysed text may come from previously structured metadata, from the actual information contained in the geographic features or from expressions incorporated by the system's user, either as annotations or as incorporated texts for queries.

Although the issue has been handled with a degree of success in communities such as the Natural Language Processing [37] and the Geographical Information Retrieval [2, 36], the issue discussed in this research work takes in other different domains, since the knowledge developed by the reference investigators is neither reproducible as yet nor integrated into the GI or the GIS.

Starting point

The work of many authors² has left a wide, robust theoretical base on the aspects to be taken into account for incorporation of time into information systems and the integration of the spatiotemporal variables into the GIS. Others³ have established the necessary concepts for the extraction and treatment of the temporal expressions and the bases for semantic analysis of the GI. In spite of the significant advances carried out, both the issue of the space-time relationship and the incorporation of the semantic analysis into the GIS are still open research fields, and more importantly, none of the authors has raised the issue as the space-time-semantics triad. There are no previous studies on this matter going beyond modelling based on cognitive analysis [50], the frameworks proposal for semantic interoperability in GIS applications [73], or the use of general semantic elements (Web Service Modelling Ontology, Internet Reasoning Service) in specific geographic settings such as emergency management [74].

A landmark in the research line in which this proposal is framed is the work presented by Gail Langran [38, 39, 41-44], who stated that a reasonable objective for the GIS would be to be capable of following up the changes occurring in a certain area by storing the historical data. She pointed out that the Information Systems in the 80's (both spatial and alphanumeric) tended since then to omit the historical record

² Langran [40, 42, 43], Worboys [85, 86, 89], Peuquet [59-61, 63], Yuan [94, 96-101], Wachowicz [83], Galton [18-23], Hornsby [24, 93], Nixon [51]

³ Pustejovsky [27, 30], Verhagen [46, 81, 82], The MITRE Corporation [9, 72], Jones [33-36], Manning [45], Markowetz [47], Mennis [49, 50]

of the data and they did not store previous versions of the system. In order to resolve this weakness, she proposed a conceptual, logical and physical model to enrich systems with temporal capabilities. She introduced the necessary concepts for the changes in geographic features to be stored cumulatively without data duplication at the time of their storage. She later presented [40] a series of annotations applicable to the spatiotemporal systems with which the first general bases of this type of systems were settled. Although the incremental model presented by Langran [43] is conceptually similar to our proposal, we will explain later on this article (Section 4) how the spatiotemporal factor $\pm \delta t$ and the semantic descriptor S determine that the proposed metamodel is different in concept, hence new and innovative. Following sections will explain in deep our proposal.

Identification of the problem

At the present time the models and data store systems of GI manage the time variable inefficiently in regard to users' needs, most particularly the possibility of carrying out analyses and a follow-up of the geographic features as a continuum. It is possible to register changes that occur in the reality that surround us (real world) and there are methods for implementation of basic temporal analyses allowing us to find out about the changes occurring within a certain period of time. Nonetheless, this task is carried out with a dataset captured at different times and states, what implies the utilisation of multiple *photos of the reality*. Every one of these photos is generated at a specific time t_n and registered independently within the system, ignoring the temporal and spatial correlation of the particular feature and the correlation of the features as a set⁴. This model known as Snapshot [1] was the first one to incorporate temporality into the spatial databases, and in spite of being so old and inefficient due to the constant duplication of data and attributes, their basic concepts still persist in the present-day systems [15, 16]. Thus, in a follow-up system one has an independently stored layer set that is kept as historical archives to which additional processes have to be applied in order to check the change zones and other characteristics inherent to the evolution of the territory. From this viewpoint, an initial finite set representing *Current Reality* (CR) is derived. We define the set CR as follows:

$$CR = \{t_0, t_1, t_2, t_3, \dots, t_n \quad / \quad t_0 \neq t_1 \dots \neq \dots, t_n \}$$

where: $t_0 \rightarrow$ Initial reality. (Photo 0)
 $t_n \rightarrow$ Reality in a subsequent time n (Photo n)

Each element t_n corresponds to a graphic output representing one or several levels of the modelled reality (layers) at a specific time. The fact that the set CR is defined by multiple elements and the number of these elements increases over time as the

⁴ The correlation of the geographic features as a set was established by Waldo Tobler – First Geography Law

changes are registered implies many drawbacks. From the viewpoint of usage and capabilities provided by the system, it is possible to mention:

- Lack of a binding historical register of the represented features.
- Users do not achieve directly (or even indirectly in some cases) the desired answers from the spatiotemporal analyses carried out.
- Inability to develop real spatiotemporal analyses. Michael Worboys [87] indicates two cases that are not resolved naturally and are related to changes in space, time and attributes: (i) changes in population density in a certain district within a fixed time interval; in this case not only the variation in the number of inhabitants is solely taken into account, but also the changes in the legal and geographic boundary line of the particular zone; (ii) the evolution of the morbidity in a fast growing city in a period of two decades.
- Attribute→space→time relationships without one-to-one matching.
- Inability to find other related levels of information or associated geographic features.
- Time of query and processing longer than really needed in queries related to historical data or temporal characteristics.

From the viewpoint of the control and register of the information status, a possible solution to these deficiencies would be the handling of the versioning, however difficulties arise such as:

- The intrinsic need to manage versions. The historical registry of spatial data is carried out following the methods developed for conventional databases [69, 70].
- Information duplication (geometries and attributes) in zones not going through changes. A difficulty derived from versioning style with which the data are currently treated within the spatial databases. As example, the GeoDataBase model of ESRI® [15].
- Lack of data versions (releases). There are evidence of the registered versions but this does not imply identifying how many versions an entity has in store; there might be one or as many as existing versions in the database.
- Possible incompatibilities between the spatial and the descriptive components, taking into account a possible update of the attributes (alphanumeric data) but not of the spatial component (geometry). Example: the census.
- Unnecessary package traffic on the computer networks derived from data duplication.

In addition to these deficiencies, we have to mention the inability of the GI to be comprehended from a semantic perspective, i.e. its meaning. The data are not prepared to interact with users in natural language and they are not ready to define neither their own context nor their meaning. This is due to the fact that the natural flow of information treatment in the GIS is User → System ← Data, the system being the one interpreting the user's commands (requests) to process subsequently the assigned user's commands taking the related data. In this scenario, everything revolves around the system by having the ability to interpret user commands and data structure; the user does not need to directly interact with the stored data, it just

proposes tasks and processes. Under these conditions the user loses self-reliance, remains assigned to a second place and sees him(her)self constrained by the own capabilities of the system. If data were better fitted semantically, able to describe their context and inform who, what and how they are, defining the set they belong to, then, they could interact among them, relate to one another and set up natural subsets (e.g. buildings, rivers, ways, trees), independent of their storage, going beyond the data model controlling them. This would help the user with procedures since the system would not have to transform every order, and part of it could be comprehended directly by the data. In this case a flow of the type $\text{User} \leftarrow \text{System} \leftrightarrow \text{Data}$ would be generated.

Description of the proposal

The above-mentioned deficiencies are indeed a problem requiring attention and needing solutions. To this effect, we have considered interesting to present a proposal improving the storage of the dynamic GI in the spatiotemporal (δt) and semantic (S) domains, with the purpose of optimising its retrieval, management, analysis and by and large the tasks based on spatial and spatiotemporal reasoning. Therefore, we propose the definition of a metamodel for storage of the dynamic GI –DGI– to be appropriate for different application domains through specific models and apt for materialisation through mark-up languages. As a result, a new robust, dynamic and flexible storage format is anticipated that integrates the spatial, the temporal and the semantic components. Based on the presented bibliographic reference framework and having exposed the issue, the starting hypothesis for definition of this research work is as follows.

The hypothesis on which this research is based on is the lack of the semantic and temporal components in the current structures of Geographic Information storage and which causes the spatiotemporal analyses to be deficient. The proposal of a new model incorporating an independent temporal structure and a semantic meaning would optimise such storage and would allow improving GI retrieval, processing and analysis capability. If this hypothesis is substantiated, the integration of the geographic, semantic and temporal components through standards would allow:

- exploiting efficiently the dynamic nature of GI;
- optimising the modelling, analysis and transfer of spatiotemporal GI;
- changing the current GI storage methods that do not appropriately fit the dynamic reality they represent;
- relying on well structured data to carry out geographic analysis taking into account the meaning of the data and the time variable;
- taking the first steps toward compatibility in the representation and analysis of data that include the time variable between the different available platforms, hence moving forward toward the semantic and temporal interoperability of the spatiotemporal data.

In order to substantiate our hypothesis, we expect to propose and implement a metamodel for the enriched storage of GI involving temporal and semantic structure, also enabling the optimization of retrieval and dynamic representation of geographic features, the ability to carry out spatiotemporal analyses, and the interaction and exploitation of data in natural language. Broadly speaking, the objective of this work is to define a model of GI storage with the following characteristics:

- Integration of the dynamic nature of GI.
- Incorporation of semantic components describing the meaning and the temporal aspect of the stored data.
- Incorporation of semantic elements enabling the user to naturally interact with them (Natural language interaction).
- Integration of spatiotemporal structures that will allow registering incrementally the changes occurred in geographic features.
- Definition of rules for interaction of any stored dataset.
- Design to be implemented with mark-up languages.

Research questions.

This research work will let to know why has not been possible to implement the spatiotemporal models proposed in the last 10 years? What challenges are involved therein? Why have not done it? In addition, we will attempt to discover if it makes sense to implement a model envisaging the semantic enrichment of data beyond their description with metadata.

Proposal and methodology

Proposal and anticipated advantages.

This work would optimise the register of geographic information through a new storing structure, incorporating the succeeding geometric and alphanumerical changes occurring over time on the geographic features, hence avoiding information duplication; furthermore, this work would permit to know the reality of a registered geographic feature at any time. We intend to provide data with semantic elements that should in turn describe them by using mainly available standards and specifications. We propose to generate a model describing a unitary set (singleton) for dynamic data storing, semantically enriched and called *Proposed Reality –PR–* defined as follows:

$$PR_0 = t_0 \cup \delta t;$$

$$PR = \{PR_0, S\}; \rightarrow PR = \{(t_0 \cup \delta t) / S \in PR \wedge \exists \text{ one and only one}\};$$

where: $t_0 \rightarrow$ Initial reality
 $\delta t \rightarrow$ Spatiotemporal change of reality (t_x)
 $\delta t \rightarrow$ Spatiotemporal factor
 $S \rightarrow$ Semantic descriptor

The spatiotemporal factor δt would make possible to know the reality (current state) of a geographic feature at a certain time and to represent the changes of the modelled object, hence of the stored feature. The changes in attributes and geometry may be registered and looked up at any point of the temporal scale in which the information has been registered; a discretisation of the model may therefore be inferred. The set contains a semantic descriptor (S) that will grant the stored data the necessary description so that data are self-describable and may relate according to their geographic and temporal nature. The following conditions are proposed for this set:

- $\forall t_x \in PR_0, \exists \rightarrow S \in PR$
- $PR_0 \subset S \quad \forall S \in PR_0$
- $t_0 = \text{dom}(t)$
- $t_x = t_{x-1} + \delta t \quad \wedge \quad t_x = t_{x+1} - \delta t$
- $\sum_{x=0}^n t_x = t_n$
- $\sum_{x=0}^n \delta t_x = \delta t_n$

This proposal could optimise the current process through the creation of a model for GI storage in which the dynamic component of the represented reality is registered, thus reducing the problems exposed for the finite set CR . With the PR model, in addition to the above-mentioned aspects in the hypothesis, it is anticipated that the outcome of this research study allow:

- Optimising the volume of storage due to the elimination of coincident geometries, hence improving transfer time of geographic data belonging to time series or collections.
- The dynamic representation of the temporal changes registered in geometry and attributes owing to the data intrinsic timeline. These changes may be linear or cyclic depending on the nature of the modelled feature.
- Optimising response time in queries and spatiotemporal analyses.
- Optimising processes applied to spatial and temporal reasoning.
- Improving the interaction and relationship of data of the same nature through semantic descriptors.
- Enabling human-data interaction in natural language.

Methodology.

This research work will be based on standard mark-up languages. Among the likely useful languages for implementation of the proposal, the following stand out: Geography Mark-up Language –GML– [54], Keyhole Mark-up Language –KML– [53], SpatialML [72], TimeML [25], DARPA Agent Mark-up Language [11], Web Ontology Language –OWL– [90], Resource Description Framework (RDF) [91], SPARQL Query Language for RDF [92], in addition to other *de facto* or *ad hoc* standards related to semantic, temporal, and/or GI storage aspects. To substantiate the defined hypothesis, and to answer the research questions, following the proposal in Figure 1, the following activities are put forward:

- Finding an efficient way of fusing together and uniting space and time in the structures of storage with mark-up languages (i.e. GML and TimeML). Here the storing methods based on mark-ups and binaries, as well as the coincidences and differences between methods and time mark-up standards will be assessed.
- Evaluation of GML-based temporal models in other specific application fields. The analysis of the temporal elements of the Aeronautical Information eXchange Model (AIXM) [17], and of the models applied to the Geological Time Systems [71] and CHRONOS [26] are proposed.
- Analysis of the appropriate form for incorporation of semantic, temporal and descriptive annotations of data. The concepts applied here will be based on widely disseminated methodologies such as the Geographic Information Retrieval [47] [14] [75] and the Natural Language Processing [45].
- Definition of the general concepts of the GI storing metamodel. Setting up the model layers and their characteristics.
- In order to validate the proposal, the metamodel will be implemented through two specific models. The implementation will be able to answer questions that have not been possible to solve so far. We will expose the deficiencies, difficulties and weak points as well as the strengths and opportunities offered.

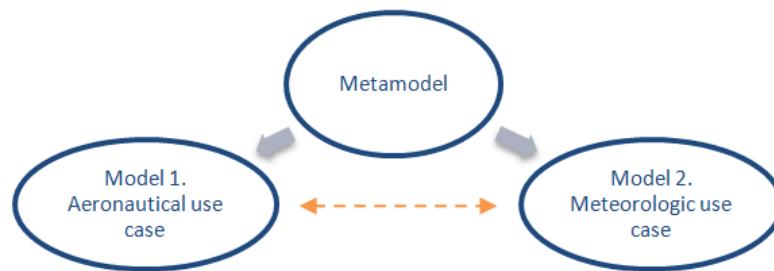


Fig. 4. Relationship between the proposed meta-model and the validation models

Conclusion and expected contributions

As a conclusion, it is possible to state that the semantic and temporal enrichment of the GI and its implementation through mark-up languages for its integration into the GI management systems is, we believe, the next natural step in the research carried out on the space-time issue in the Information Systems. It is necessary to carry on with the work of Peuquet, Langran, Armstrong, Snodgrass, Worboys, Yuan, Wachowicz, Galton, Hornsby, Frank, Jones, Shen, and all the researchers who have contributed some element to lay the foundations of the change in the paradigm toward the temporal analysis of GIS and the semantic interpretation of the GI. This proposed research work is warranted since it is anticipated to make progress, (i) setting up mark-up languages as an integrating element of the spatial, temporal and semantic elements; (ii) adding further descriptions to data so that they will be able to interact, to describe themselves and to provide the system with temporal and spatiotemporal

dynamism as well as meaning. Temporal dynamism through a robust mark-up language; spatiotemporal dynamism by identifying incremental variations; enhancement of meaning by identifying the represented feature type; and (iii) developing concepts enabling progress in geosemantics.

The innovation of this research work lies in the proposal of a metamodel for representation, retrieval, reasoning and spatiotemporal and semantic analysis of GI. The anticipated contributions to be gained from this research are:

- Proposal of a metamodel enabling exploitation of the GI dynamic component.
- Definition of the method for incorporation of the semantic component into the GI storage structure.
- Definition of the method for integration of an independent temporal component into the GI.
- Proposal of a new format for the GI integrating the temporal, spatial, and attribute components, providing a semantic-temporal support.
- Proof of concept through implementation of mark-up language-based storing spatiotemporal structures.

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